Emergent $\mathcal{PT}$ Symmetry and Exceptional Points in a Driven Dirac Impurity

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A stationary rotating drive tunes the coalescence of two decaying impurity resonances in a Hermitian model with a finite-band radial spin–orbit-coupled bath and remote auxiliary levels. The exact auxiliary rotation supplies coherent spin-odd coupling and modifies the bath's SOC-induced relative decay. Retaining the full frequency dependence and both open helicity cuts, we establish the exceptional point (EP) numerically through a nonlinear double zero, a rank-one inverse kernel and pole exchange. Retuning the controls gives an EP family as SOC varies, with positive physical decay rates. The result concerns the complete causal resonance problem; exact parity–time symmetry is confined to a normalized constant-core benchmark. Interactions are then added to the same microscopic model. An exact finite-chain self-energy provides an approximate continuum embedding with matching pole topology at two chain lengths in a nearly empty regime. A continuum Kotliar–Ruckenstein saddle also admits EPs at and near mean filling one, where substantial double occupancy identifies a charge-fluctuating state. These results establish finite-chain and mean-field continuations at their respective levels; a continuum-converged local-moment EP remains unresolved.

I. INTRODUCTION

We ask whether a drive acting on remote bath orbitals can tune a coalescence of decaying impurity resonances when finite-band dispersion and the complete causal frequency dependence are retained. Driven impurities connect this question to local interactions and state preparation. Keldysh methods provide a framework for their response [1–3]; related approaches address dissipative impurities and open spin systems [4, 5]. Exceptional points (EPs), at which eigenvalues and eigenvectors coalesce, have been studied in boundary-sensitive bands [6], interacting atoms [7], dissipative Hubbard models [8], and photonic, acoustic and diffusive systems [9–15]. Condensed-matter observations are reported in Ref. [16].

For correlated electrons, lattice coherence [17], driven exchange [18], nonequilibrium DMFT [19] and real-time impurity solvers [20] motivate studying how a drive changes local resonances. Exactly solved boundary and impurity models [21–23], non-Hermitian RG and causality studies [24–26], and NRG, cold-atom and slave-boson approaches [27–29] address complementary interaction questions. Floquet multichannel Kondo couplings [30] and dissipative Luttinger liquids [31] are distinct settings.

Other settings include complex-hybridization Kondo breakdown [32], exact high-bias and time-dependent impurity limits [33, 34], and Hermitian equilibrium $\mathcal{PT}$ protection in $CeCo_2P_2$ [35]. A companion mathematical work [36] develops the controlled linearized contact model; no integrability of the complete driven kernel is claimed here.

Here we study a defined radial SOC bath with an impurity and driven remote levels. Its stationary rotating protocol produces a tunable coherent spin-odd term that competes with the bath's SOC-induced relative decay. The full-momentum result is an EP of two analytically continued impurity resonance poles. Exact $\mathcal{PT}$ symmetry is used only for a normalized constant-core benchmark; it is not assumed for the full frequency-dependent kernel. The distinction also separates the retarded pole calculation from the thermal closure used in supporting Keldysh checks.

Projection of a Hermitian system-plus-continuum Hamiltonian to a non-Hermitian resonance problem, including impurity EPs, is established [37, 38]. Our contribution is the explicit connection between the stationary drive and a family of passive resonance EPs in the specified finite-band bath. The same bath determines the coherent coupling and relative decay, so the double-zero and pole-exchange tests use its full analytic continuation. This establishes the coalescence beyond the normalized constant-core approximation and identifies the real controls needed to reach it.

The interaction calculations keep this stationary-drive Hamiltonian and ask how the resonance pair continues after adding U. Exact finite-chain responses supply a self-energy for an approximate continuum embedding; a continuum slave-boson saddle independently tests filling and retuning within mean field. The occupied saddle EP is charge fluctuating, so neither mean filling one nor a large ratio $U/\bar{\Gamma}_0$ is identified with Kondo strong coupling. Symmetry-protected interacting EPs are known in other models [39]; the separate spin-symmetric impurity control in the Supplemental Material tests the role of symmetry without replacing the SOC bath that generates the present

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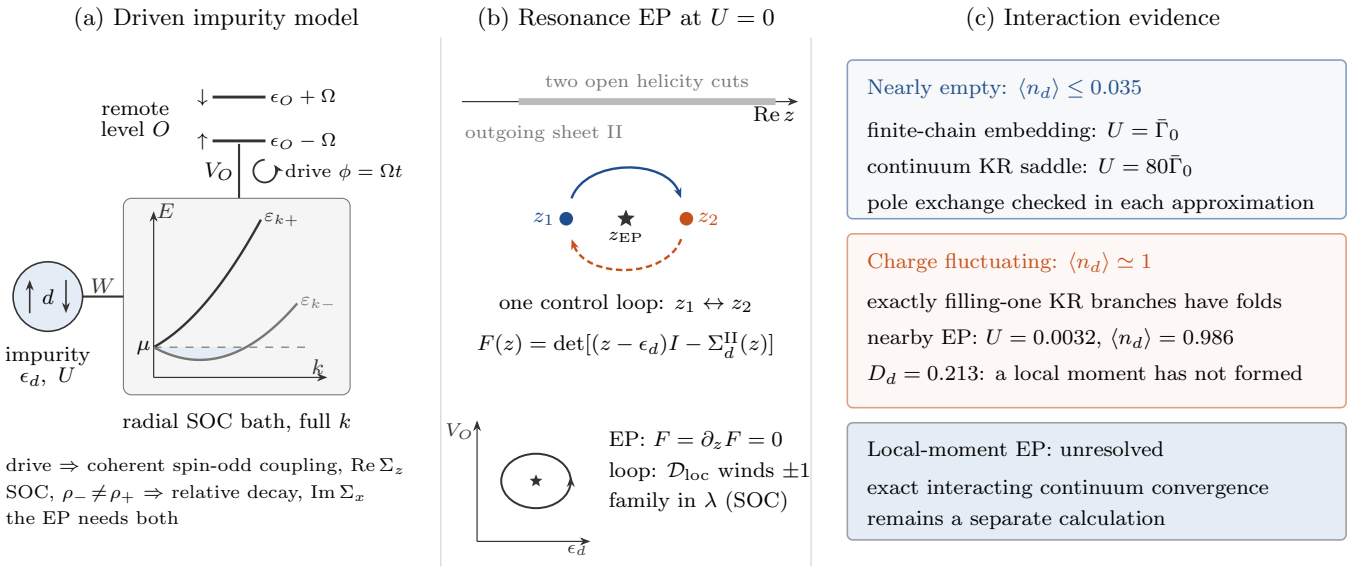


FIG. 1. **Model, pole EP and interaction evidence** (schematic). (a) The impurity d couples through W to the full radial SOC bath, which couples through V_O to a remote fermionic level. Uniform phase rotation shifts the auxiliary energies by $\mp\Omega$. (b) The outgoing-sheet inverse kernel has a rank-one double zero; encircling it in two real controls exchanges the poles. (c) The finite-chain embedding and continuum KR saddle give method-specific interaction evidence. Here $\bar{\Gamma}_0 = W^2/4 = 0.0025$; at $U = 0$ the retarded EP is independent of filling. The nearly occupied KR points remain charge fluctuating, and their constrained folds do not exclude nearby EPs. This panel is not a phase diagram.

mechanism. Phenomenological non-Hermitian impurity and RG constructions address related questions [25, 26, 28]. The normalized-core dynamics, thermal-closure checks and restricted contact algebra help interpret the pole result; they are supporting analyses of separately stated limits.

Figure 1 summarizes the model, the resonance EP and its interaction continuations. Section II defines the model. Section III contains the central causal-pole calculation and finite-U tests. Sections IV–IX supply the supporting effective-Hamiltonian, mean-field, response and scattering analyses.

II. MODEL AND HAMILTONIAN

The Hermitian model is $H = H_{\mathrm{bath}} + H_{\mathrm{imp}} + H_{\mathrm{imp\text{-}bath}} + H_O + H_{\mathrm{hyb}}[\phi(t)]$: an Anderson impurity, a radial two-component bath, remote fermionic orbitals and driven bath–orbital hoppings. The linear-in-k splitting motivates "Dirac-like"; the quadratic curvature and cutoff are retained. The Supplemental Material [40] collects notation and parameters.

A. Bath

We define a radial effective bath, with the two-dimensional measure $k\,dk/(2\pi)$ and fixed-axis SOC:

$$H_{\mathrm{bath}} = \sum_{k,\sigma} \epsilon_k\, c^\dagger_{k\sigma} c_{k\sigma} + \lambda \sum_k k\big(c^\dagger_{k\uparrow} c_{k\downarrow} + \mathrm{h.c.}\big), \tag{1}$$

where $c^\dagger_{k\sigma}$ creates a bath electron with radial momentum k (in units of inverse lattice constant) and spin $\sigma \in \{\uparrow,\downarrow\}$; $\epsilon_k = \alpha_{\mathrm{b}} k^2$ ($\alpha_{\mathrm{b}} > 0$; $\alpha_{\mathrm{b}} = 1$ in all numerical calculations), with radial cutoff $0 \leq k \leq k_{\max} = \pi/4$; and $\lambda > 0$ is the spin-orbit coupling. The separate energy cutoff D_{uv} enters only logarithmic and scaling formulas below. The SOC term in Eq. (1) uses one spin axis for every momentum direction. It is therefore odd under time reversal and is not the Rashba coupling of a time-reversal-symmetric C_{3v} host: for such a host and a local $m = 0$ orbital, C_3 forces the self-energy of the impurity spin doublet to be diagonal, and no spin-mixing channel arises. Equation (1) is accordingly an effective radial channel model. An orbital impurity coupling to a suitable equal-C_3-character pair is one possible realization to investigate; neither that construction nor a host-specific projection is established here. The

chiral rotation $c_{k\pm} = (c_{k\uparrow} \pm c_{k\downarrow})/\sqrt{2}$ diagonalizes $H_{\rm bath}$ exactly, yielding SOC-split chiral dispersions $\varepsilon_{k\pm} = \alpha_{\rm b} k^2 \pm \lambda k$ with $\varepsilon_{k+} \neq \varepsilon_{k-}$ for all $k \neq 0$. For the full radial dispersions, the two positive-energy roots give

$$\begin{aligned} \rho_+(\omega) &= \frac{\Delta - \lambda}{4\pi\alpha_{\rm b}\Delta}, \qquad \rho_-(\omega) = \frac{\Delta + \lambda}{4\pi\alpha_{\rm b}\Delta}, \\ \Delta &= \sqrt{\lambda^2 + 4\alpha_{\rm b}\omega}. \end{aligned} \tag{2}$$

whenever both roots lie inside the radial cutoff. Hence $\rho_+ + \rho_- = 1/(2\pi\alpha_{\rm b})$ for $\omega \to 0^+$: the complete quadratic-plus-SOC radial bath is not pseudogapped. The familiar $|\omega|/\lambda^2$ law describes only the small-k cone patch of one branch. Below zero energy the lower branch instead has a band minimum at $-\lambda^2/(4\alpha_{\rm b})$ and the associated radial van Hove edge. The constant *sum* makes the spin-even width flat, while the channel difference $\rho_- - \rho_+ = \lambda/(2\pi\alpha_{\rm b}\Delta)$ remains; in the full radial kernel this static difference, not the drive, supplies the relative decay (Sec. III). The full-k calculation below retains this exact structure; the symmetric linear-DOS form is used only as an explicitly idealized analytic benchmark.

B. Impurity

The local Anderson impurity is $H_{\rm imp} = \sum_\sigma \epsilon_\xi\, d_\sigma^\dagger d_\sigma + U\, n_\uparrow n_\downarrow$, where $d_\sigma^\dagger$ creates an impurity electron with spin σ, $n_\sigma = d_\sigma^\dagger d_\sigma$ is the occupation, ϵ_ξ is the bare impurity level before the projected bath-induced detuning is included, and $U > 0$ is the on-site Coulomb repulsion.

C. Impurity–Bath Hybridization

The impurity couples directly to the $m = 0$ (s-wave) bath channel via

$$H_{\text{imp-bath}} = \sum_{k,\sigma} \Big(W_k\, c_{k,0,\sigma}^\dagger d_\sigma + \text{h.c.} \Big), \tag{3}$$

where W_k is the bare hybridization amplitude. W_k is time independent and independent of the frozen drive control β_0. At the $U \to \infty$ slave-boson saddle the physical spinon–bath vertex is rW_k, with $r = |b_c|$; in the finite-U saddle of Sec. III F the scalar r becomes a spin-dependent matrix R.

The drive reaches the impurity only through the auxiliary dressing of this bath channel (Sec. III).

D. Auxiliary Angular-Momentum Modes

As motivation for the auxiliary sector, a scalar trigonal term $\beta_3(k) k^3 \cos 3\theta$ (where $\beta_3(k)$ is the momentum-dependent trigonal crystal-field strength, distinct from the hybridization amplitude β_0 introduced below; and θ is the in-plane momentum angle) couples the $m = 0$ (s-wave) bath channel to higher angular-momentum harmonics $m = \pm 3, \pm 6, \ldots$ via the C_3 selection rule $\Delta m = \pm 3$. These higher harmonics are represented by auxiliary fermions $O_{k,m,\sigma}$ with angular momentum index $m \neq 0$:

$$\begin{aligned} H_O = &\sum_{k,m\neq 0,\sigma} \epsilon_O^{(m)}\, O_{k,m,\sigma}^\dagger O_{k,m,\sigma} \\ &+ \sum_{\substack{k,m,\sigma \\ m,m+3\neq 0}} \beta_3(k) k^3 \big(O_{k,m+3,\sigma}^\dagger O_{k,m,\sigma} + \text{h.c.} \big), \end{aligned} \tag{4}$$

where the mode energies satisfy $\epsilon_O^{(m)} = \epsilon_O^{(-m)}$ as a chosen symmetry of the auxiliary crystal-field sector. This does not restore time reversal of the fixed-axis bath. The fields $O_{k,m,\sigma}$ provide remote off-shell levels ($\epsilon_O^{(m)} \gg D_{\rm uv}$) and are integrated out exactly within the specified quadratic model.

E. Periodic Drive and Hybridization

The periodic drive couples the physical bath ($m = 0$) to the remote higher angular harmonics. We write the Hermitian bath–auxiliary hybridization as

$$H_{\rm hyb}[\phi(t)] = \sum_{k,\sigma,m\neq 0} \left[V_m(k) e^{i\zeta_\sigma \phi(t)} c^\dagger_{k,0,\sigma} O_{k,m,\sigma} + \text{h.c.}\right]. \tag{5}$$

Here $\zeta_\uparrow = +1$ and $\zeta_\downarrow = -1$ label the spin projection, and $V_m(k)$ is the chosen coupling from the radial channel to the auxiliary sector, motivated by the scalar C_3 selection rule and nonzero for $m = \pm 3$ at this order; higher harmonics are reached through H_O. An angle-resolved vertex $V_m e^{i3m\theta} c^\dagger_{k,\theta,\sigma}$ would instead couple O_m to the bath harmonic $3m \neq 0$, which has no overlap with the impurity channel. The full-k calculation keeps a single remote level of this form, with vertex V_O and energy ϵ_O. The normalized frozen-kernel scans use the slow oscillatory protocol $\phi(t) = (\pi/4)\sin(\Omega t)$. In that protocol the passive control is the instantaneous angular velocity $\dot\phi(t_{\rm av}) = (\pi\Omega/4)\cos(\Omega t_{\rm av})$; the numerical parameter β_0 used below should be read as the corresponding effective frozen-kernel amplitude, not as the value of the phase itself. Hermiticity fixes the second vertex to carry the complex-conjugate phase. Therefore the normal Gaussian contraction of a number-conserving auxiliary sector contains phase differences, not phase sums. This is important: the off-shell auxiliary sector does not by itself generate anti-Hermitian gain or loss.

The unitary change of variables $O^{\rm lab}_{k,m,\sigma} = e^{-i\zeta_\sigma\phi(t)} O^{\rm rot}_{k,m,\sigma}$ removes the phase from Eq. (5). The vertices become phase-free and the auxiliary energies acquire the Berry shift $\epsilon^{(m)}_O - \zeta_\sigma \dot\phi(t)$. Thus a gapped Hermitian auxiliary sector produces a real spin-odd coherent shift. For the oscillatory protocol this statement is an adiabatic low-energy projection. For the stationary rotating protocol $\phi(t) = \Omega t$, however, this energy shift is constant and the rotated quadratic Hamiltonian is exactly time independent; no freezing or gradient expansion in Ω is then required. In this frame the drive is equivalent to a static spin-dependent splitting $\mp\Omega$ of the remote level, and its role is to make that splitting tunable. Only the remote fields are rotated, so the conduction and impurity operators are unchanged by this transformation. Their occupations still depend on the state preparation; a static Hamiltonian does not fix them. The passive non-Hermitian component appears only in the second step, when this shifted channel is embedded in the retarded Dirac bath. For the oscillatory protocol the projection requires $|V|/\epsilon_O \ll 1$ and $\Omega \ll \epsilon_O$. A full cycle sweeps the two signs of the frozen control; nonadiabatic EP traversal and energy pumping are outside the scope of this description.

III. COARSE GRAINING AND EMERGENT $\mathcal{PT}$ SYMMETRY

A. Keldysh Path Integral and Angular-Mode Integration

We work within the Keldysh formalism [1], writing the partition function as a coherent-state path integral on the closed time contour. The bath and auxiliary angular modes enter quadratically and can be eliminated at the Gaussian level. The local Coulomb interaction on the impurity is not treated in this Gaussian step; it is treated below by self-energy embedding from exact finite-chain responses (Sec. III E) and by a finite-U slave-boson saddle of the continuum bath (Sec. III F). The separate $U \to \infty$ slave-boson approximation in Sec. V is a supporting constant-kernel benchmark.

Integrating out the auxiliary fields gives the influence kernel $\Xi_\sigma = V_\sigma g_O V^\dagger_\sigma$. Because the auxiliary sector conserves particle number, the drive phases enter only as differences, and the Wigner transform shifts the auxiliary self-energy by $\zeta_\sigma\dot\phi$ (Supplemental Material).

To leading order in the slow drive and in the off-shell expansion, the auxiliary-dressed $m = 0$ bath therefore has the form $\Sigma^{\rm bath,R}_\sigma = \Sigma^{\rm bath,R}_0 + \zeta_\sigma \Delta_{\rm aux} + O[(\Omega/\epsilon_O)^2]$. Here $\Delta_{\rm aux}$ is real in the low-energy window. Microscopically one may write, schematically,

$$\begin{aligned}\Delta_{\rm aux}(\omega, t_{\rm av}) = &-\dot\phi(t_{\rm av})\,\partial_{\epsilon_O} \sum_{m\neq 0} |V_m|^2 \\ &\times \mathcal{P}\frac{1}{\omega - \epsilon^{(m)}_O},\end{aligned} \tag{6}$$

up to form-factor coefficients.

The bare impurity–bath hybridization W_k in Eq. (3) remains time independent. The drive reaches the impurity through the dressed bath propagator. In contour notation the two Gaussian eliminations give

$$G_{c,\sigma}^{-1} = g_c^{-1} - \Xi_\sigma, \qquad \Xi_\sigma = V_\sigma \circ g_O \circ V_\sigma^\dagger, \tag{7}$$

$$\Sigma_{d,\sigma} = W^\dagger \circ G_{c,\sigma} \circ W, \tag{8}$$

with $\circ$ the contour convolution; the drive enters only through the path $d \to c \to O \to c \to d$.

In the frozen off-shell window, write $\Xi_\sigma^R = \Xi_0^R + \zeta_\sigma \delta\Xi^R$ with $\delta\Xi^R \in \mathbb{R}$. Expanding the dressed-bath resolvent gives the general first-order correction

$$\delta\Sigma_d^R = W^\dagger G_{c,0}^R \, \delta\Xi^R \, G_{c,0}^R W. \tag{9}$$

Although the auxiliary correction is real, the retarded continuum propagators are complex, so the transmitted impurity correction has both coherent and dissipative parts. Defining $\Sigma_{d,0}^R(\omega) = \Lambda(\omega) - i\bar{\Gamma}(\omega)$, we write $\Sigma_{d,\sigma}^R = \Sigma_{d,0}^R + \zeta_\sigma \Delta_{\rm coh} + i\zeta_\sigma \Gamma_{\rm PT}$. For the nearly rigid scalar shift used in the analytic estimates, $\delta\Xi^R \simeq \Delta_{\rm aux}$ and the spin-even auxiliary background varies slowly. Then

$$\Sigma_{d,\sigma}^R(\omega, t_{\rm av}) \simeq \Sigma_{d,0}^R(\omega - \zeta_\sigma \Delta_{\rm aux}) \simeq \Sigma_{d,0}^R - \zeta_\sigma \Delta_{\rm aux} \partial_\omega \Sigma_{d,0}^R, \tag{10}$$

which gives $\Delta_{\rm coh} = -\Delta_{\rm aux} \partial_\omega \Lambda$ and $\Gamma_{\rm PT} = \Delta_{\rm aux} \partial_\omega \bar{\Gamma}$. Equation (9) is the general first-order result; Eq. (10) is its rigid-shift limit. All functions are evaluated in the active low-energy window. For an SOC matrix bath, however, the exact result need not reduce to the rigid shift of the spin-even scalar self-energy. In the chiral basis set $a = (\omega^+ - \varepsilon_{k+})^{-1}$, $b = (\omega^+ - \varepsilon_{k-})^{-1}$, so that $g_0 = (a+b)/2$ and $g_x = (a-b)/2$. Then $-\partial_\omega g_0 = g_0^2 + g_x^2$ whereas $g_b \sigma_z g_b = (g_0^2 - g_x^2)\sigma_z = ab\,\sigma_z$. With the sign convention of Eq. (6), the exact first-order relative coefficient is therefore

$$\delta\Sigma_z^R = \Delta_{\rm aux} \sum_k |W_k|^2 a_k b_k,$$

$$\Gamma_{\rm PT} = \Delta_{\rm aux} \left[\partial_\omega \bar{\Gamma} - 2 \, {\rm Im} \sum_k |W_k|^2 g_{x,k}^2 \right]. \tag{11}$$

Reversing the definition of the spin-odd auxiliary shift reverses both sides of Eq. (11), without changing any rate eigenvalue. Two limits must be distinguished. For a scalar energy-dependent width ($\partial_\omega \bar{\Gamma} \neq 0$, $g_x = 0$), Eq. (11) reduces to the rigid-shift result following Eq. (10) and the drive generates a relative decay. For the quadratic radial dispersion with constant $W_k = W$ and both helicity roots open, both branches have radial velocity Δ, so ${\rm Im} \sum_k |W_k|^2 a_k b_k = 0$: the total width is flat, ${\rm Im} \sum_k |W_k|^2 g_{x,k}^2$ vanishes as well, and the first-order relative decay is zero. The cancellation holds to all orders in the drive while both channels are open (Supplemental Material). In the full-k kernel the drive therefore enters as a coherent spin-odd coupling, and the relative decay originates in the SOC helicity imbalance $\gamma = W^2\lambda/(4\alpha_{\rm b}\Delta) = \pi W^2(\rho_- - \rho_+)/2$ at zero drive. Auxiliary dressing also changes the spin-mixing decay through the on-shell roots; the cancellation of the spin-odd decay does not mean that all dissipative coefficients are drive independent.

B. Passive $\mathcal{PT}$ Core and Causal Detuning

The projected impurity self-energy is decomposed into a common bath part, a passive relative decay imbalance, and a Kramers–Kronig-related coherent detuning: $\Sigma_{\rm imp}^R = (\Lambda - i\bar{\Gamma})\mathbb{I} + (\Delta_{\rm coh} + i\Gamma_{\rm PT})\sigma_z$. The scalar term is a common Lamb shift and common decay. It is retained in the bath-controlled FDR, but at a fixed frequency it shifts both matrix eigenvalues equally. Its frequency dependence is retained when locating resonance poles. The spin-odd terms have different symmetry status: $i\Gamma_{\rm PT}\sigma_z$ is the passive relative decay imbalance, while $\Delta_{\rm coh}\sigma_z$ is a real coherent detuning generated by the same causal retarded bath self-energy.

The SOC bath supplies the spin-mixing channel: a Hermitian principal-value part and, through the helicity imbalance, an anti-Hermitian part. In the original spin basis, the local bath Green function has the matrix structure $g_b^R = [\omega^+ - \epsilon_k - \lambda k \sigma_x]^{-1} = g_0 \mathbb{I} + g_x \sigma_x$, with $g_x(k,\omega) = \frac{\lambda k}{(\omega^+ - \epsilon_k)^2 - \lambda^2 k^2}$. Equivalently, $g_x = (a-b)/2$ with the two chiral propagators defined above, and the channel-odd self-energy is

$$\Sigma_x = \tfrac{1}{2}(\Sigma_+ - \Sigma_-) = \tfrac{1}{2}(\Lambda_+ - \Lambda_-) - \tfrac{i}{2}(\Gamma_+ - \Gamma_-). \tag{12}$$

The principal-value part of this SOC off-diagonal bath dressing gives a real coherent spin-mixing coefficient, $\Delta_{\rm eff}(\omega) = \sum_k |W_k|^2 \, {\rm Re}\, g_x(k,\omega)$, renormalized in the interacting problem by the slave-boson saddle point.

The drive-induced shift cannot leak into this mixing channel at leading order: expanding the dressed bath propagator to first order in the spin-odd shift $\Delta_{\rm aux}\sigma_z$ gives $g_b^R\,\sigma_z\,g_b^R = \left(g_0^2 - g_x^2\right)\sigma_z$, because $\{\sigma_x,\sigma_z\}=0$. For the scalar-shift benchmark, omitting the additional SOC loss imbalance and subtracting the common damping defines the relative kernel

$$h_{\rm rel} = \Delta_{\rm eff}\sigma_x + (\Delta_{\rm coh} + i\Gamma_{\rm PT})\,\sigma_z. \tag{13}$$

The passive-$\mathcal{PT}$-symmetric core is obtained at the compensated detuning point $\Delta_{\rm coh}=0$:

$$h_{\mathcal{PT}} = \Delta_{\rm eff}\sigma_x + i\Gamma_{\rm PT}\sigma_z, \qquad \sigma_x h_{\mathcal{PT}}^{*}\sigma_x = h_{\mathcal{PT}}. \tag{14}$$

The real term $\Delta_{\rm coh}\sigma_z$ is not discarded. It is $\mathcal{PT}$-odd under the same operation and therefore acts as a causal detuning floor. Exact $\mathcal{PT}$ symmetry with $P=\sigma_x$ is a property of this normalized core. The complete kernel also contains the imaginary spin-mixing part $-i(\Gamma_+ - \Gamma_-)\sigma_x/2$ of Eq. (12), which is $\mathcal{PT}$-odd under $P=\sigma_x$. In the exact radial kernel, where $\Gamma_{\rm PT}=0$, the roles are exchanged: the leading-order dimer is $\Delta_{\rm coh}\sigma_z + i\gamma\sigma_x$ with $\gamma=(\Gamma_- - \Gamma_+)/2$, which is $\mathcal{PT}$-symmetric with $P=\sigma_z$, and the real part of Σ_x plays the role of the detuning in Eq. (15) below. The certified pole EP is the nonlinear-pole version of this coalescence and does not rely on exact $\mathcal{PT}$ symmetry. The squared half-splitting of the frozen matrix Eq. (13) is $s^2 = \Delta_{\rm eff}^2 + (\Delta_{\rm coh} + i\Gamma_{\rm PT})^2$. For real $\Delta_{\rm eff}$, $\Delta_{\rm coh}$, and $\Gamma_{\rm PT}$, exact coalescence in the passive sector requires

$$\Delta_{\rm coh}\Gamma_{\rm PT} = 0, \qquad \Delta_{\rm eff}^2 + \Delta_{\rm coh}^2 - \Gamma_{\rm PT}^2 = 0. \tag{15}$$

Thus, for a nonzero relative decay imbalance, the exact passive EP occurs at the compensated point $\Delta_{\rm coh}=0$ and $|\Delta_{\rm eff}| = |\Gamma_{\rm PT}|$. Away from the compensated point, the Kramers–Kronig detuning prevents a literal pole coalescence. Minimizing $|s|$ over real $\Delta_{\rm eff}$ at fixed $\Delta_{\rm coh}$ and $\Gamma_{\rm PT}$ gives $|s|_{\rm min} = \sqrt{2|\Delta_{\rm coh}\Gamma_{\rm PT}|}$ when $|\Delta_{\rm coh}| \le |\Gamma_{\rm PT}|$; otherwise the minimum occurs at $\Delta_{\rm eff}=0$ and equals $\sqrt{\Delta_{\rm coh}^2 + \Gamma_{\rm PT}^2}$. For the idealized sharp-cutoff benchmark the floor is already 0.90–$1.00\,\Gamma_{\rm PT}$ one linewidth from the compensation energy, so this frozen finite-band benchmark shows avoided coalescence (Supplemental Material).

The rigid-shift estimate gives $\Gamma_{\rm PT} \sim \Omega|V|^2\epsilon_O^{-2}\partial_\omega\bar{\Gamma}$; reaching the normalized benchmark ratio $\Gamma_{\rm PT}/\bar{\Gamma} = 0.10/0.12$ would require $\Omega/\epsilon_O \gtrsim 1.061(\epsilon_O/|V|)^2$, incompatible with an off-shell quasistatic drive. Supplemental Fig. S1 therefore displays the constant-core geometry only.

C. Stationary rotating-drive resonance EP

The preceding negative statement is specific to the normalized benchmark and the oscillatory quasistatic protocol. For $\phi(t)=\Omega t$, the rotating-frame auxiliary levels are shifted by the constant $\mp\Omega$. Keeping their exact frequency dependence, the curved SOC dispersion, and the complete radial momentum integral, define on the outgoing resonance sheet

$$F(z;\epsilon_d,V_O) = \det\left[(z-\epsilon_d)\mathbb{I} - \Sigma_d^{\rm II}(z;V_O)\right]. \tag{16}$$

Here z is complex energy, ϵ_d denotes the bare impurity gate in this full-bath calculation, and II labels the outgoing resonance sheet (SM App. B). A nonlinear pole EP is a double zero $F=\partial_z F=0$ with $\partial_z^2 F \neq 0$ and a rank-one inverse kernel. To certify its existence without evaluating a singular eigenbasis, let $z_c(\epsilon_d,V_O)$ be the continuously branch-tracked solution of $\partial_z F(z_c)=0$ on a closed control loop C, and define $\mathcal{D}_{\rm loc} = F(z_c;\epsilon_d,V_O)$ with winding $W_C = \Delta_C \arg\mathcal{D}_{\rm loc}/(2\pi)$. Assume that the tracked critical point extends as one smooth branch over the enclosed control domain and that $\partial_z^2 F \neq 0$ there. The implicit-function theorem guarantees such a neighborhood locally around a nondegenerate critical point; a boundary check alone does not establish the extension over an arbitrary domain. Regarding $\mathcal{D}_{\rm loc}$ as a map from the two real controls to $(\mathrm{Re}\,\mathcal{D}_{\rm loc}, \mathrm{Im}\,\mathcal{D}_{\rm loc})$, its boundary winding is the Brouwer degree whenever $\mathcal{D}_{\rm loc}\neq 0$ on C. A nonzero degree forces at least one enclosed zero (with regular zeros counted by their Jacobian sign). At such a zero, $F=\partial_z F=0$; the nonzero second derivative makes it a quadratic pole coalescence, and the rank-one inverse-kernel test excludes a scalar degeneracy. Explicit pole exchange and the half-winding of their separation provide the equivalent EP2 monodromy check. The full two-cut calculation in the Supplemental Material gives a representative point for $\lambda=0.05$, $W=0.10$, $k_{\rm max}=\pi/4$, $\epsilon_O=120$ and $\Omega=60$:

$$\begin{aligned} V_O &= 1.21342201, \qquad \epsilon_d = 0.01517035, \\ z_{\rm EP} &= 0.01197283 - 0.00280722i. \end{aligned} \tag{17}$$

Both helicity cuts are open and the pole lies $1.351|\,\mathrm{Im}\,z_{\rm EP}|$ above the driven $k=0$ threshold; the local degree is $+1$, and residuals, rank test, passivity margin (0.761), and off-shell ratios are tabulated in the Supplemental Material. At

$\omega = \mathrm{Re}\, z_{\mathrm{EP}}$ on the physical sheet the drive enters as the coherent coupling $\mathrm{Re}\,\Sigma_z = 6.50 \times 10^{-4}$, the spin-mixing loss is $\mathrm{Im}\,\Sigma_x = 5.96 \times 10^{-4}$ (5.57×10^{-4} for $V_O = 0$), and $\mathrm{Im}\,\Sigma_z$ vanishes, as required with both channels open; the common half-width is $\bar{\Gamma} = 0.0025$. The EP is therefore drive controlled: without the drive the coherent coupling is absent. The certificate is a temperature-independent retarded-kernel statement. The estimate $T_{\mathrm{b}} \lesssim 10^{-3}$ is a linewidth-resolution scale for thermally broadened spectroscopy, not a critical temperature for the $U = 0$ EP.

The off-shell condition is $|V_O|/|\epsilon_O \mp \Omega - \omega| \ll 1$ throughout the active energy window, with witness ratio 0.0202. Exact shifted denominators are retained; no slow-drive expansion is used. Retuning U, gate or chemical potential requires rechecking this window, open channels and positive rates. Stationarity alone does not establish a thermal preparation.

Figure 2 distinguishes the control loop from the complex-energy trajectories. The pole midpoint $(z_1 + z_2)/2$ moves along the control loop, so the individual energy trajectories need not enclose the fixed central value z_{EP}. The invariant tests are the winding of $\mathcal{D}_{\mathrm{loc}}$ and the exchange of the tracked poles, equivalently the half-winding of $z_1 - z_2$.

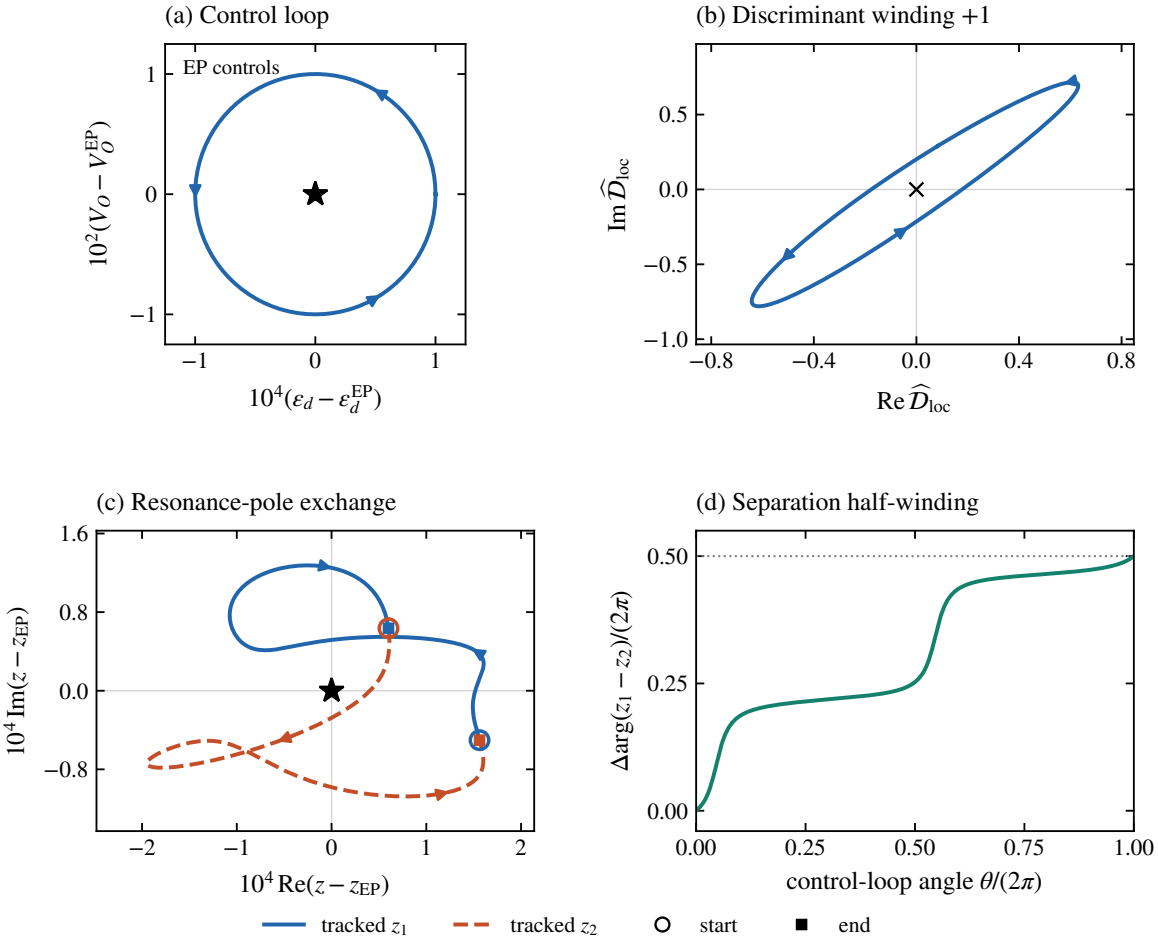


FIG. 2. **Full-momentum two-cut nonlinear causal-pole certificate.** The stationary rotating-drive witness uses $\lambda = 0.05$ and fixed $W = 0.10$, $K = \pi/4$, $\epsilon_O = 120$, $\Omega = 60$, and the solution in Eq. (17). (a) The control ellipse encloses the EP controls; its half-widths are 10^{-4} in ϵ_d and 10^{-2} in V_O. (b) $\hat{\mathcal{D}}_{\mathrm{loc}} = \mathcal{D}_{\mathrm{loc}}/\max_C |\mathcal{D}_{\mathrm{loc}}|$ winds once around zero. (c) Actual pole displacements from z_{EP} (star), with arrows showing increasing control angle. Open circles and filled squares mark the exact initial and final points; the colours exchange at each endpoint. These curves are pole trajectories, not bath branch cuts, and need not individually enclose the star. (d) The continuously unwrapped separation phase advances by π in one circuit, giving half-winding $+1/2$. All pole solves pass the separation and residual guards. The certificate concerns the full curved kernel, not its constant-core approximation.

D. Parameter continuation and real-frequency diagnostic

Continuation in SOC, with W held fixed at 0.10 while the real gate and the off-shell auxiliary vertex are retuned by the same double-zero equations, gives a finite locus over $0.02 \le \lambda \le 0.1578$ (Fig. 3). Independent local-winding and pole-exchange certificates pass at $\lambda = 0.05$, 0.10, and 0.15. The continued pole approaches an open-channel threshold near the upper end while the rate matrix remains passive. The endpoint is not classified; approach of $\mathrm{Re}\, z_{\mathrm{EP}}$ to a real threshold alone does not establish a complex branch-point collision. The isolated parameter values quoted above are therefore points on a one-parameter family of causal pole EPs in the full curved kernel, as expected for a codimension-two condition.

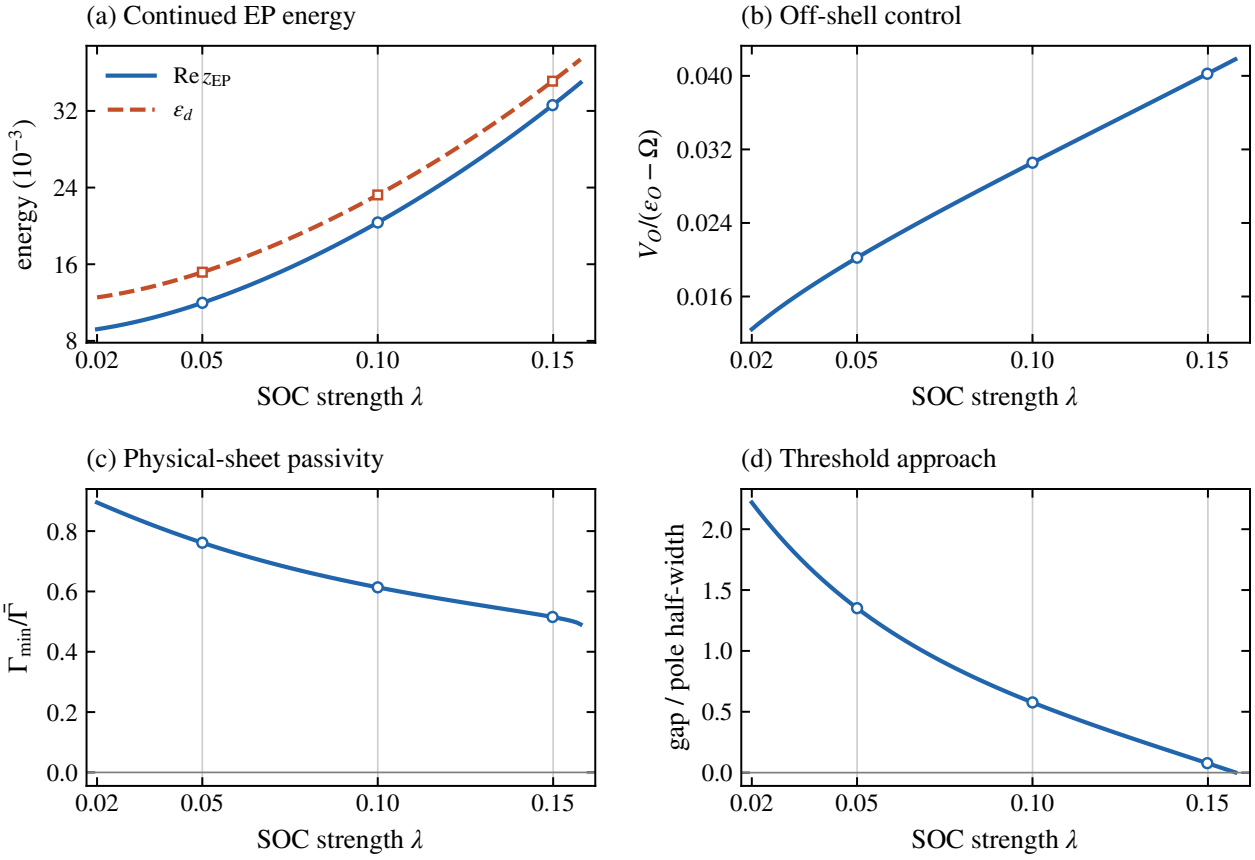


FIG. 3. **SOC family of full-momentum nonlinear pole EPs.** The double-zero equations are continued in λ with $W = 0.10$ fixed, while the real impurity gate ϵ_d and the off-shell auxiliary vertex V_O are retuned. Vertical guides mark $\lambda = 0.05$, 0.10, and 0.15, where independent local-winding and pole-exchange certificates give winding $+1$, pole permutation (12), and half-winding $+1/2$. The off-shell ratio remains below 0.042 and the physical rate matrix remains positive. The last panel shows the gap $(\mathrm{Re}\, z_{\mathrm{EP}} - \omega_{\mathrm{th}}^{+})/|\,\mathrm{Im}\, z_{\mathrm{EP}}|$ to the driven upper $k = 0$ threshold. This is a one-dimensional locus in the enlarged parameter space; neither its threshold endpoint nor a two-dimensional EP region is claimed.

Figure 4 gives a complementary real-frequency diagnostic. With ϵ_d and V_O fixed at the $\lambda = 0.10$, $\Omega = 60$ certificate, choose the first scalar reference root ω_* of $\mathrm{Re}[\omega_* - \epsilon_d - \Sigma_0^R(\omega_*)] = 0$ in the stated search window. The retarded upper-rim self-energy defines $s^2 = \Sigma_x^2 + \Sigma_z^2$, and the color shows $C_s = \mathrm{Re}\, s^2/|s^2|$. Positive and negative values indicate predominantly real and imaginary frozen-matrix splitting, respectively. Since s^2 is generally complex, the line $C_s = 0$ is a crossover, not an EP boundary or an exact $\mathcal{PT}$ transition. Diagonalizing this kernel at ω_* does not locate its complex resonance poles. The marked EP is independently obtained from $F = \partial_z F = 0$. As Ω varies, $B_{\mathrm{raw}}(0)$ is subtracted anew: the slice holds the recentered bath reference fixed, equivalent to adjusting the corresponding scalar bath offset along with the drive.

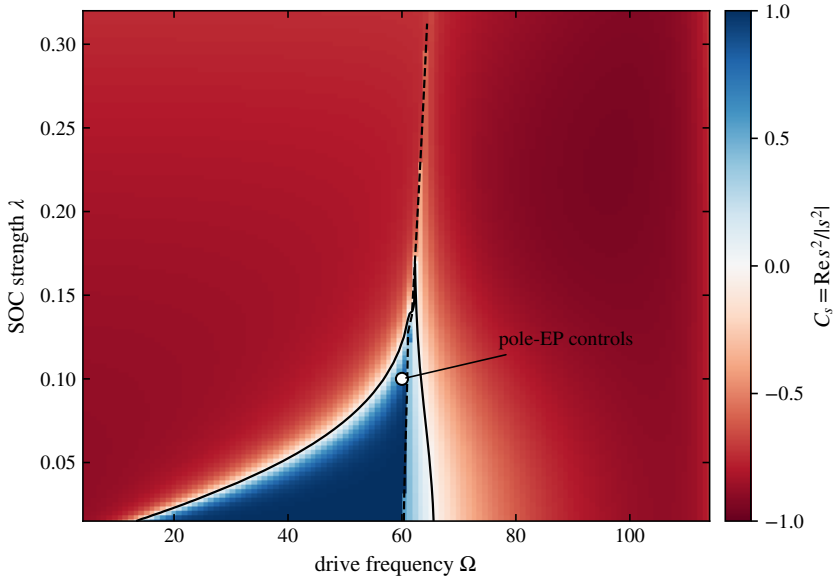


FIG. 4. **Retarded real-frequency splitting diagnostic.** Color: $C_s = \mathrm{Re}(\Sigma_x^2 + \Sigma_z^2)/|\Sigma_x^2 + \Sigma_z^2|$ at the first scalar reference root $\omega_* \in (10^{-4}, 0.60)$. The matrix is evaluated at $\omega_* + i\eta$, $\eta = 10^{-10} > 0$; upper-rim convergence and positive half-widths are checked separately. Parameters are $U = 0$, $W = 0.10$, $K = \pi/4$, $\epsilon_O = 120$, $\epsilon_d = 0.02323413$ and $V_O = 1.83318448$. Solid: $C_s = 0$ crossover. Dashed: change in the number of open channels at ω_*. The circle marks controls of an independently certified complex-pole EP. Neither line defines a region of EPs or exact $\mathcal{PT}$ symmetry.

E. Finite-U self-energy embedding

We next test whether the representative $\lambda = 0.05$ causal pole EP survives a local Anderson interaction without imposing the double-zero condition on the interacting self-energy. The positive 2×2 matrix spectral density of the same rotating full-k bath is logarithmically discretized with $\Lambda = 1.5$, mapped by block Lanczos to a matrix Wilson chain, and averaged over four z shifts. We first use 12-site chains to locate the finite-U center and evaluate the zero-temperature correction-vector Green functions G and F in exact fixed-charge sectors and form the equation-of-motion estimator (averaging the chain responses before taking the matrix ratio)

$$\Sigma_U(z) = UF(z)G^{-1}(z). \tag{18}$$

Solving $F_U = \partial_z F_U = 0$ for the determinant of the full analytic two-cut kernel dressed by Eq. (18) gives, at $U = 0.0025$,

$$\begin{aligned} z_{\mathrm{EP}}^{(U)} &= 0.01177670 - 0.00225815i, \\ \epsilon_d^{(U)} &= 0.01440607, \qquad W^{(U)} = 0.09036305. \end{aligned} \tag{19}$$

The complex poles belong to this approximate continuum-kernel embedding; the closed Hermitian finite chain has real eigenvalues. Exact charge-sector responses do not establish the continuum limit or the outgoing analytic continuation of the interacting self-energy. These are separate from the exact two-cut continuation of the noninteracting bath. Here U is 4×10^{-3} of the bandwidth $\alpha_{\mathrm{b}} k_{\max}^2 = 0.617$, while $U/\bar{\Gamma}_{\mathrm{self}} = 1.225$ with $\bar{\Gamma}_{\mathrm{self}} = (W^{(U)})^2/4 = 0.00204$. The level lies about $7\bar{\Gamma}_{\mathrm{self}}$ above the chemical potential, so the impurity is nearly empty. Direct $T = 0$, $\mu = 0$ checks on the same four-shift $N = 12$ reference give $\langle n_d \rangle = 0.017064$ and $\langle n_{d\uparrow} n_{d\downarrow} \rangle = 5.73 \times 10^{-7}$ (Supplemental Material); the continuum saddle of Sec. III F gives 0.024 and 1.9×10^{-6} at its own retuned EP. The calculation probes a weakly correlated impurity resonance and does not establish a local-moment or Kondo regime. Testing an occupied regime requires independent pole retuning and bath-convergence checks; near-single occupancy and suppressed double occupancy alone do not establish either an EP or Kondo screening. Section III F addresses both points with a saddle that is evaluated on the continuum and on these chains. Additional occupied-bath tests find stable charge observables but bath-dependent spin density and self-energy (SM App. C); they do not certify a continuum local-moment EP.

Figure 5 gives the topological gate: on a small control ellipse in (ϵ_d, W) the branch-tracked local discriminant has winding -1, the two poles exchange, and all guards pass; z-average, kept-space, and sum-rule checks are given in the Supplemental Material. The sign differs from the $U = 0$ certificate because the degree depends on the chosen pair of controls and its orientation; only its magnitude, one, is fixed.

As an independent chain-length gate, the center is reoptimized separately on the $N = 10$ and $N = 12$ complete-basis chains. Their maximum relative coordinate drift is 3.746×10^{-3}. On a shared ellipse with half-widths 2×10^{-4} in ϵ_d and 2×10^{-3} in W, both chains have local-discriminant winding -1, endpoint pole exchange, and pole-separation half-winding $-1/2$. At 256 contour points,

$$\frac{\max_{j=1,\ldots,256} |D_{12,j} - D_{10,j}|}{\min_{j=1,\ldots,256} |D_{10,j}|} = \frac{1.638 \times 10^{-9}}{2.417 \times 10^{-9}} = 0.6775 < 1, \tag{20}$$

which supports a nonvanishing straight-line homotopy on the resolved contour. A uniform version of this inequality, together with a smooth critical-point branch over the enclosed domain, would preserve the Brouwer degree. The numerical maximum is not a bound between samples. The 128-to-256 refinement and the independently tracked pole exchanges provide finite-chain stability evidence, without bounding the omitted chain tail.

The two chain lengths hold one and two electrons in their ground states, so the occupied part of the bath is represented coarsely. At $U = 0$ the 12-site chains give $\langle n_d \rangle = 0.0207$, against 0.0347 for the continuum. The $N = 10 \to 12$ comparison therefore tests the topology but not this occupied-bath resolution, and the coordinates in Eq. (19) are specific to the chains; Sec. III F compares them with a distinct continuum mean-field approximation.

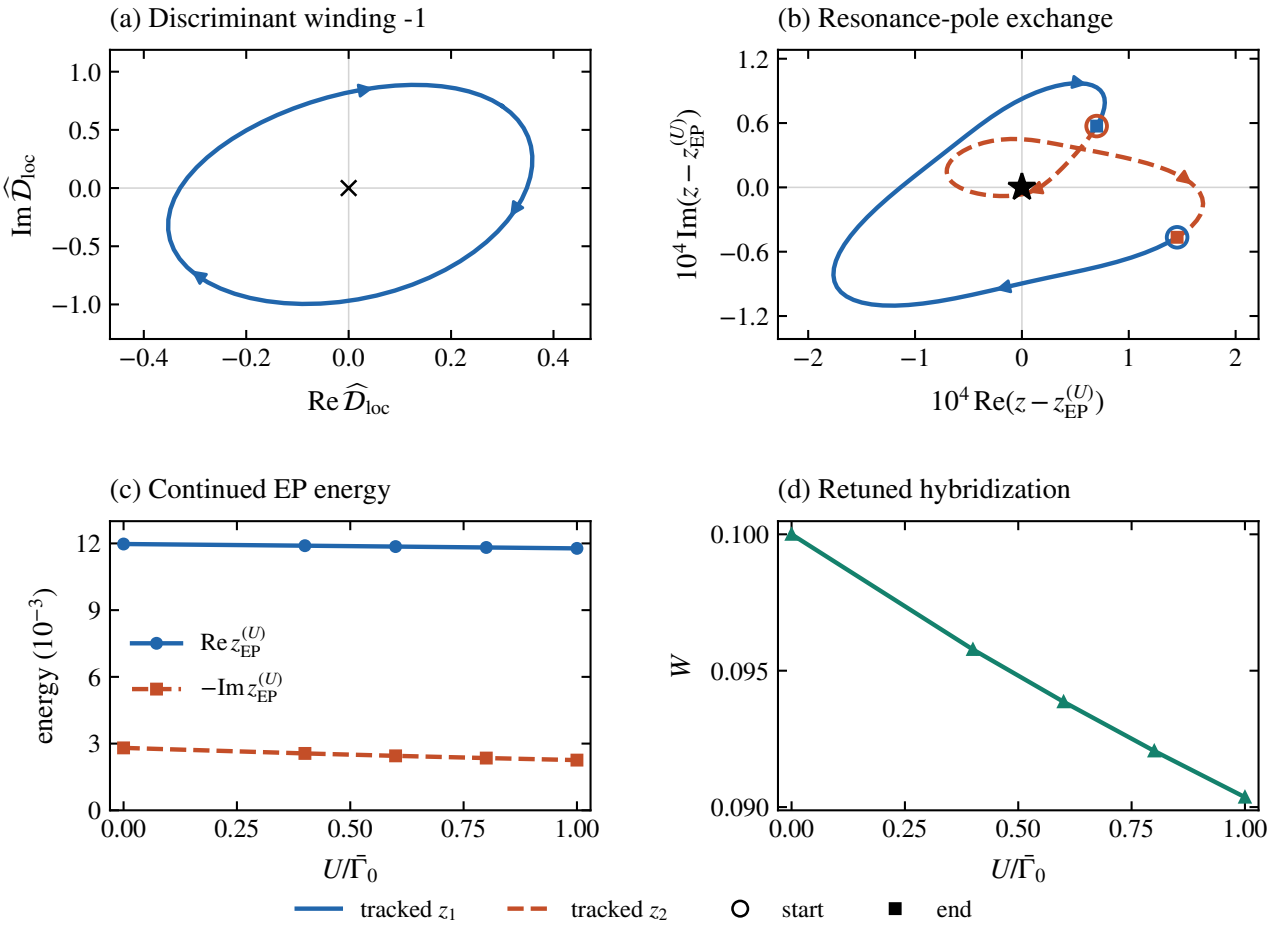


FIG. 5. **Finite-U continuation of the causal full-k pole EP.** (a) The branch-tracked local discriminant winds once clockwise around the origin on the interacting control ellipse at $U = 0.0025$. (b) The two nearby poles exchange after one circuit; open circles and filled squares mark the initial and final tracked labels, and the star marks Eq. (19). (c) The center of the finite-chain self-energy embedding varies smoothly from $U = 0$ to $U/\bar{\Gamma}_0 = 1$, where $\bar{\Gamma}_0 = 0.0025$. (d) The retuned impurity hybridization W on the same path. The discriminant in (a) is normalized by its maximum modulus on the loop; the displacement scale in (b) is 10^{-4}. Kept-space errors relative to the exact response on the same four shifted 12-site chains are tabulated in SM App. C; they are not used to establish the winding. The separate $N = 10 \to 12$ common-contour gate is given by Eq. (20) and detailed in the Supplemental Material.

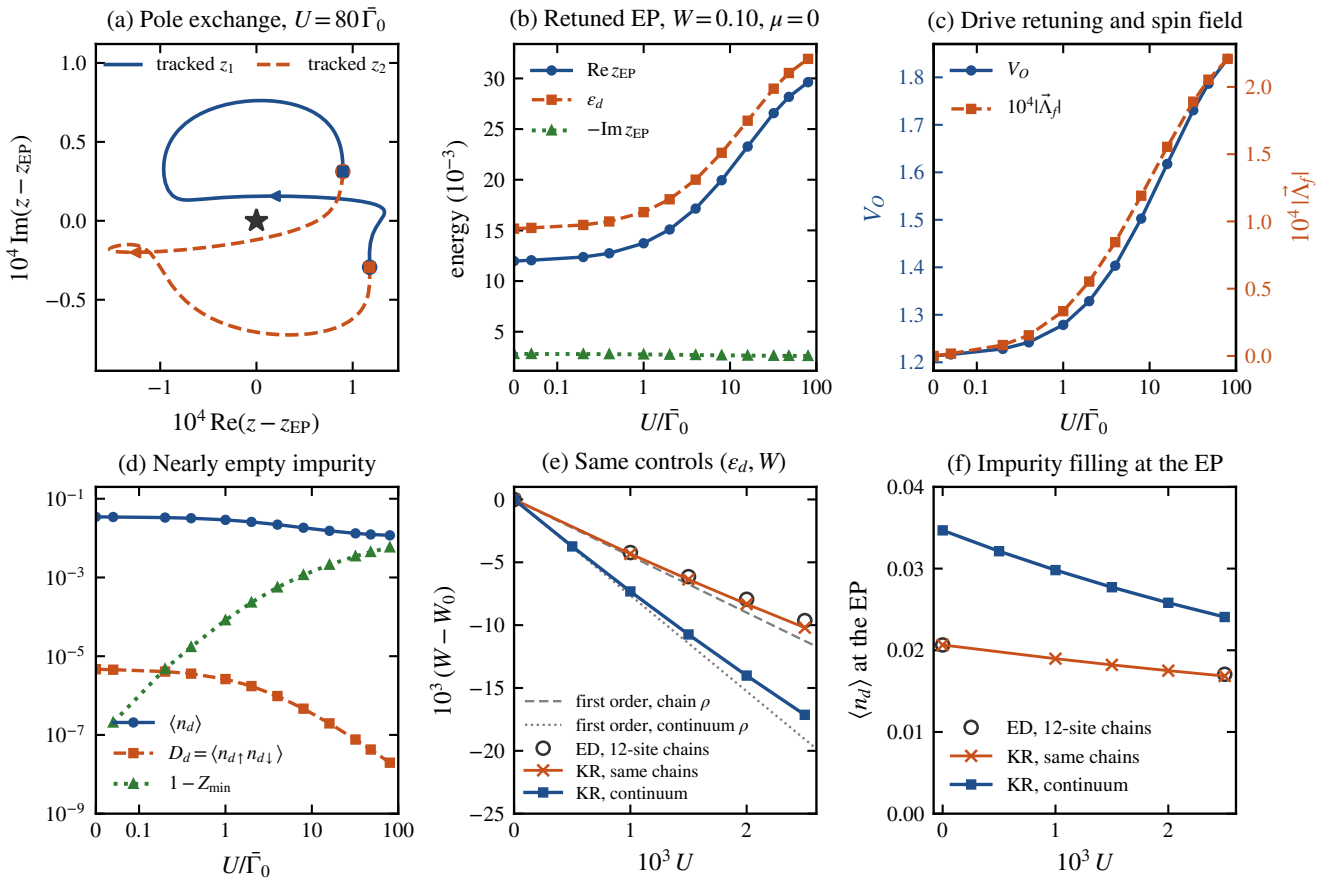


FIG. 6. **Finite-U continuation of the full-k resonance EP with the continuum Kotliar–Ruckenstein saddle** ($T=0$). (a) Pole exchange at $U=0.2=80\bar{\Gamma}_0$ ($\bar{\Gamma}_0=0.0025$, $\mu=0$) on the control ellipse of Fig. 2, with the saddle re-solved at each of 128 points. Open circles and filled squares mark the initial and final poles; the colours exchange at each endpoint, and the star marks z_{EP}. (b) EP energy and gate on the branch with $W=0.10$ fixed and (ϵ_d, V_O) retuned; $U=0$ is Eq. (17). (c) Retuned drive vertex and the traceless constraint field $|\vec{\Lambda}_f|$ that moves the EP. (d) Impurity filling, double occupancy and smallest quasiparticle weight. (e) Shift of W for the controls of Sec. III E, (ϵ_d, W) at fixed V_O: ED on the four 12-site chains (the path of Eq. (19)), KR with mesh-averaged inverse-kernel corrections on those chains, and KR on the continuum. Lines are the first-order Hartree–Fock rates with the chain and continuum density matrices. (f) Impurity filling at the EP for the same three calculations. Panels (e,f) compare method and bath dependence; the different ED and KR mesh averages are specified in SM App. N. All KR results are mean-field saddle results. The filling-one EP and its constrained continuation are examined separately in Fig. 7.

F. Finite-U continuation on the continuum bath

The finite-chain embedding establishes the numerical pole topology at the tested chain sizes; convergence of the occupied bath remains unresolved. To separate bath discretization from the interaction approximation, we next use a continuum-bath saddle. The interaction enters the double-zero condition only through its self-energy near the coalescing poles. The spin-even part shifts both poles and is absorbed by the gate, while the traceless part adds to the spin-odd and spin-mixing channels of Eq. (16) and moves the double zero. Because the $U=0$ EP has local degree ±1 (Sec. III C), a retuned EP persists, under the same smooth-branch assumption, while the interaction keeps $\mathcal{D}_{\mathrm{loc}}\neq 0$ on the certifying loop. Globally, this argument gives no protection once a discriminant zero crosses the loop, or the smooth critical-point branch or chosen resonance sheet ceases to exist. Interaction-induced changes of EP structure are possible [39]. At first order in U the self-energy is the static Hartree–Fock field $\Sigma_{\mathrm{HF}}=U[(\mathrm{Tr}\,\rho)I-\rho]$, with $\rho_{\alpha\beta}=\langle d^\dagger_\beta d_\alpha\rangle=\rho_0 I+\vec{\rho}\cdot\vec{\sigma}$. Its traceless part $-U\vec{\rho}\cdot\vec{\sigma}$ is an exchange field along the impurity polarization. With the continuum ρ of the $U=0$ witness it fixes the initial retuning, $dW/dU=-7.63$ and $d\epsilon_d/dU=-0.551$ at fixed V_O.

Beyond first order we use the finite-U Kotliar–Ruckenstein (KR) slave-boson saddle [41] in its spin-rotation-covariant matrix form [42], at $T=0$ and with the preparation of Sec. III E. A renormalization matrix R and a real constraint

field $\Lambda_f = \Lambda_{f0} I + \vec{\Lambda}_f \cdot \vec{\sigma}$ enter the coherent propagator $G_d(z) = R\,[zI - \Lambda_f - R\,\Sigma_d^{\rm II}(z)\,R]^{-1} R$. The bath kernel is integrated along the imaginary axis without discretization, and $F_{\rm KR} = \det\big[zI - \Lambda_f - R\,\Sigma_d^{\rm II} R\big] = \partial_z F_{\rm KR} = 0$ is solved together with the saddle, which is re-solved whenever a control changes (SM App. N). Since $R = I + O(U^2)$ and $\Lambda_f = \epsilon_d I + \Sigma_{\rm HF} + O(U^2)$, the saddle reproduces the first-order self-energy; its retuning rates agree with Hartree–Fock to better than 1%. Quadrature refinement and independent functional derivatives support numerical convergence within mean field; SM App. N reports residuals, local-stability checks and pole-loop refinement.

At $\mu = 0$, with $W = 0.10$ fixed and (ϵ_d, V_O) retuned [Fig. 6(a)–(d)], the EP branch extends to $U = 0.2 = 80\bar{\Gamma}_0$ ($\bar{\Gamma}_0 = 0.0025$), or 0.32 of the bandwidth. There $z_{\rm EP} = 0.02963 - 0.00261i$, $\epsilon_d = 0.03193$ and $V_O = 1.8389$, with both helicity channels open, positive rates and $V_O/(\epsilon_O - \Omega) = 0.031$; the retuning is of order one but saturates. The impurity stays nearly empty, $\langle n_d\rangle = 0.0118$ with $Z \geq 0.994$, while double occupancy is suppressed to $D_d = 2.0 \times 10^{-8}$, 0.09 of $\det n$. The EP is moved by the traceless field $|\vec{\Lambda}_f| = 2.2 \times 10^{-4}$, a third of the $U = 0$ coherent coupling $\mathrm{Re}\,\Sigma_z$. At weak U this field is the exchange part of $\Sigma_{\rm HF}$; at strong U it is a spin-dependent kinematic shift, the level of one spin being pushed by the blocked hybridization of the other, a mean-field counterpart of Haldane's level renormalization [43] in a polarized bath. At $U/\bar{\Gamma}_0 = 8$ and 80 the loop of Fig. 2, with the saddle re-solved at 128 points, gives winding $+1$, pole exchange (12) and half-winding $+1/2$ [Fig. 6(a)]; all three are unchanged at 256 intervals.

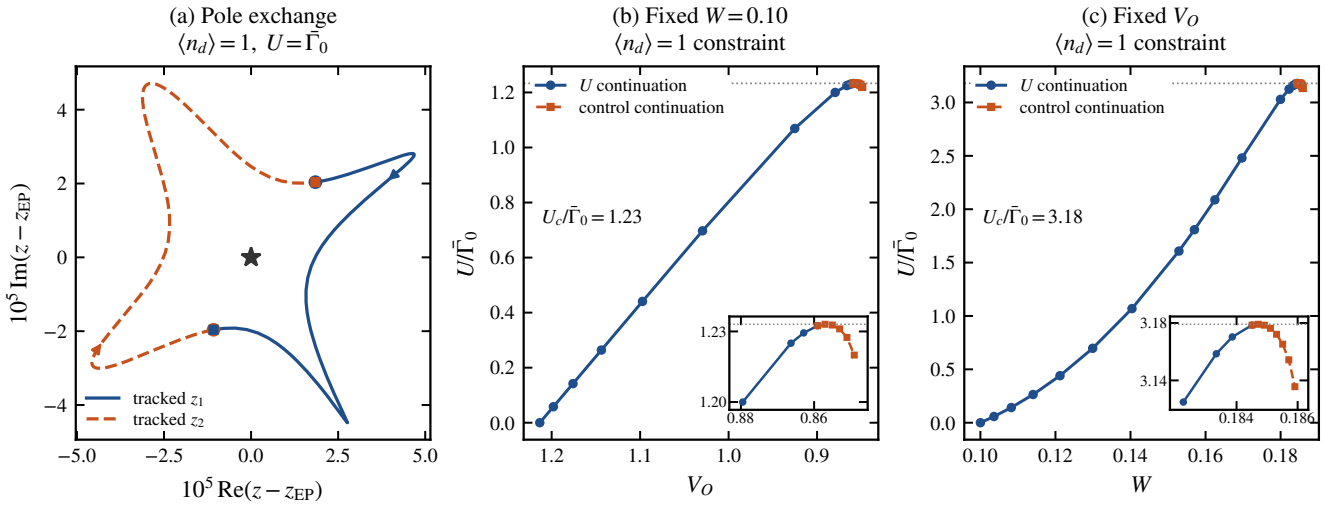


FIG. 7. **An occupied, charge-fluctuating EP and its constrained continuation in the continuum KR saddle** ($T = 0$). (a) Pole exchange at $\langle n_d\rangle = 1$ and $U = 0.0025 = \bar{\Gamma}_0$. The star marks $z_{\rm EP}$; open circles and filled squares mark the initial and final poles. At the center, $D_d = 0.229$ and $P_1 = 0.542$, so this is not a local-moment state. The loop keeps μ fixed and re-solves the saddle; its controls and numerical checks are given in SM App. N. (b,c) EP branches constrained to $\langle n_d\rangle = 1$, with $W = 0.10$ or V_O fixed, respectively. Circles continue in U; squares continue in the indicated control with U as an unknown. Insets resolve local folds at $U_c/\bar{\Gamma}_0 = 1.23$ and 3.18. Here $\bar{\Gamma}_0 = 0.0025$ is the fixed reference width, including in (c) where W varies. These constrained turns do not exclude nearby EPs: releasing the filling constraint yields the verified KR point $U = 0.0032$, $\langle n_d\rangle = 0.98587$ (not plotted; SM App. N). All results are mean-field saddle results.

Figure 6(e,f) compares ED and KR results on the four 12-site chains with the continuum KR saddle. The chain calculations use different mesh averages: ED forms the self-energy from averaged responses, whereas KR averages the individual inverse-kernel corrections (SM App. N). Their shifts differ by 2.5–5.8% in W and 1.7–4.3% in ϵ_d over the tabulated interval. These are observed differences, not an isolated method-error bound. The continuum KR shift of W is 1.68–1.69 times its chain value, consistent with the chain's smaller occupied density matrix. The independently certified continuum KR point at $U = 0.0025$ has $W = 0.082865$, $\epsilon_d = 0.013902$ and $z_{\rm EP} = 0.011684 - 0.001874i$, with winding -1 and pole exchange. No empirical correction of these coordinates is identified as an exact continuum interacting result.

G. Resonance EP at mean filling one

To test whether the mean-field EP is confined to a nearly empty impurity, we also retune the chemical potential to obtain $\langle n_d\rangle = 1$. Figure 7(a) shows pole exchange at $U = 0.0025 = \bar{\Gamma}_0$, with the saddle re-solved along the control loop. At this EP, $D_d = \langle n_{d\uparrow} n_{d\downarrow}\rangle = 0.229$ against $\det n = 0.241$, and the single-occupancy probability is

$P_1 = \langle n_d \rangle - 2D_d = 0.542$. Thus mean filling one supports an EP within the continuum KR approximation, but this state is charge fluctuating: the mean occupation alone does not establish a local moment.

Keeping the filling exactly one gives local folds at $U_c = 0.003083$ when (ϵ_d, V_O, μ) are retuned with $W = 0.10$ fixed, and $U_c = 0.007948$ when (ϵ_d, W, μ) are retuned with V_O fixed [Fig. 7(b,c)]. These are turns of constrained EP branches, not boundaries of an EP-free phase. Fixing μ at the first fold and releasing the exact-filling constraint gives a locally stable EP at $U = 0.0032 > U_c$, $\langle n_d \rangle = 0.98587$ and $D_d = 0.21330$. Its double-zero and rank-one checks, winding -1 and pole exchange are reported in SM App. N. This nearby point demonstrates why the filling constraint matters; with $P_1 = 0.55928$, it also remains charge fluctuating.

SM Fig. S5 follows a separate occupied path without an EP search. A local-moment EP remains unresolved, motivating further work on polarized-bath Kondo physics [44, 45].

IV. NORMALIZED HAMILTONIAN BENCHMARK AND EMERGENT NON-HERMITICITY

A. Projected Coefficients and Auxiliary Benchmark

At the slave-boson saddle (Sec. V), with gauge-invariant amplitude $r = |b_c|$, the impurity–bath vertex is rW. Since an impurity self-energy contains two spinon–bath vertices, the physical projected parametrization is $\Gamma_{\rm PT} = r^2\Gamma^{(0)}_{\rm PT}(\beta_0)$, $\Delta_{\rm eff} = r^2\Delta^{(0)}_0$ and $V_{\rm eff} = rW$. Here the superscript (0) denotes the frozen bath projection before slave-boson renormalization. The core discriminant therefore carries a common factor r^4, so the local core EP is exactly invariant under changes of $r > 0$ (Supplemental Material). This invariance follows from a renormalization common to both spins, as in the $U \to \infty$ saddle of Sec. V; it does not protect the EP location. A finite-U saddle gives spin-dependent quasiparticle factors and a traceless constraint field, which move the EP (Sec. III F). Supplemental Fig. S1 uses $r = 1$ and the normalized path $\Delta^{(0)}_{\rm eff} = 0.50$, $\Gamma^{(0)}_{\rm PT} = |\beta_0|$; the scattering, saddle and flow checks use the archived fixed-W path $\Gamma_0 = 0.12$, $\Delta^{(0)}_{\rm eff} = 0.075 + 0.050\beta_0$ and $\Gamma^{(0)}_{\rm PT} = 0.20\beta_0$ (hence $\Delta^{(0)}_{\rm eff} = 0.10$ at $\beta_0 = 0.50$). The normalized geometry has symmetric crossings at $\beta_0 = \pm 0.50$; the small-width audit uses only the positive window $0.35 \le \beta_0 \le 0.55$, with a crossing at $\beta_0 = 0.50$.

To display the constant-core geometry, we independently define a four-component auxiliary benchmark at $\Delta_{\rm coh} = 0$, in the basis $(d_\uparrow, d_\downarrow, c_{k+}, c_{k-})$. Its local coefficients and explicit two-level bath block are assigned independently; coefficients already obtained by eliminating the physical bath must not be inserted again together with that same bath. This benchmark is not the physical Dyson equation of Sec. III:

$$H_{\rm ss}(k) = \begin{pmatrix} \tilde\epsilon_\xi + i\Gamma_{\rm PT} & \Delta_{\rm eff} & V_{\rm eff}/\sqrt{2} & V_{\rm eff}/\sqrt{2} \\ \Delta_{\rm eff} & \tilde\epsilon_\xi - i\Gamma_{\rm PT} & V_{\rm eff}/\sqrt{2} & -V_{\rm eff}/\sqrt{2} \\ V_{\rm eff}/\sqrt{2} & V_{\rm eff}/\sqrt{2} & \varepsilon_{c,+}(k) & 0 \\ V_{\rm eff}/\sqrt{2} & -V_{\rm eff}/\sqrt{2} & 0 & \varepsilon_{c,-}(k) \end{pmatrix}. \tag{21}$$

Here $\tilde\epsilon_\xi = \epsilon_\xi + \delta_c + \delta_\Sigma$ is the renormalized impurity level. The Lagrange multiplier δ_c enforces the slave-boson constraint, while δ_Σ is the real projected one-body shift from the principal-value self-energy in Sec. III B (and may be absorbed into ϵ_ξ by convention). The chiral bath dispersions are $\varepsilon_{c,\pm}(k) = \alpha_{\rm b}k^2 \pm \lambda k$.

For real coefficients, the antiunitary symmetry is $P_4\mathcal{K}$, with $P_4 = {\rm diag}(\sigma_x, 1, -1)$ and complex conjugation $\mathcal{K}$; the bath-channel sign is required to preserve the hybridization block.

B. Diagonal-Hybridization Rotation

The hybridization block of $H_{\rm ss}$ has Hadamard structure $(V_{\rm eff}/\sqrt{2})\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$. The unitary rotation $U_2 = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ acting on the impurity subspace defines chiral spinon modes $\tilde d_\pm = (d_\uparrow \pm d_\downarrow)/\sqrt{2}$ and diagonalizes the hybridization block exactly. In the $(\tilde d_+, \tilde d_-, c_{k+}, c_{k-})$ basis,

$$\tilde H(k) = \begin{pmatrix} \tilde\epsilon_\xi + \Delta_{\rm eff} & i\Gamma_{\rm PT} & V_{\rm eff} & 0 \\ i\Gamma_{\rm PT} & \tilde\epsilon_\xi - \Delta_{\rm eff} & 0 & V_{\rm eff} \\ V_{\rm eff} & 0 & \varepsilon_{c,+}(k) & 0 \\ 0 & V_{\rm eff} & 0 & \varepsilon_{c,-}(k) \end{pmatrix}. \tag{22}$$

Here $\tilde d_+$ ($\tilde d_-$) hybridizes only with c_{k+} (c_{k-}), while the internal $i\Gamma_{\rm PT}$ term can still mix the two auxiliary channels. The frozen impurity eigenvalues are $E_\pm = \tilde\epsilon_\xi \pm s$, $s = \sqrt{\Delta^2_{\rm eff} - \Gamma^2_{\rm PT}}$, on either path of Sec. IV A. The plotted eigenvalues

and condition numbers use the full 4×4 kernel in Eq. (22). Eliminating its bath block gives

$$\mathcal{P}(z,k) = \mathfrak{p}_+(z,k)\mathfrak{p}_-(z,k) + \Gamma_{\rm PT}^2 = 0, \tag{23}$$

where $\mathfrak{p}_\eta(z,k) = z - \tilde{\epsilon}_\xi - \eta\Delta_{\rm eff} - V_{\rm eff}^2/[z - \varepsilon_{c,\eta}(k)]$ for $\eta = \pm$. The full quadratic-kernel EP condition is $\mathcal{P}(z_{\rm EP},k) = \partial_z\mathcal{P}(z_{\rm EP},k) = 0$. At $k = 0$ the bath energies coincide, so with $\zeta(z) = z - \tilde{\epsilon}_\xi - V_{\rm eff}^2/z$ this reduces to $\mathcal{P}(z,0) = \zeta(z)^2 - \left(\Delta_{\rm eff}^2 - \Gamma_{\rm PT}^2\right)$. Thus the impurity-related double root occurs at $\Delta_{\rm eff}^2 = \Gamma_{\rm PT}^2$ in this common-shift limit; SOC-resolved finite-k asymmetry generally unfolds it. The declared nonzero-k scan finds no additional interior EP because every minimizer lies at its upper boundary. This factorization concerns the auxiliary benchmark. Its individual eigenvalues are not identified with the full continuum resonance poles, which are obtained from Eq. (16).

V. SLAVE-BOSON MEAN FIELD AND KELDYSH SELF-CONSISTENCY

The effective impurity problem originates from a correlated Anderson level with local repulsion U. We formulate the constrained low-energy sector using the $U \to \infty$ large-N slave-boson mean-field saddle [29, 46, 47]. This section tests a separately defined passive Markov kernel. It supplies neither the finite-U self-energy of Sec. III E nor the finite-U saddle of the full curved bath in Sec. III F. Because a single boson renormalizes both spins equally, this saddle only relabels $W \to rW$ and $\epsilon_\xi \to \tilde{\epsilon}_\xi$: it changes which bare controls realize a given mean-field EP when an admissible saddle exists. This mapping does not guarantee a finite saddle at arbitrary parameters or an unchanged complex pole in the full frequency-dependent bath.

A. Slave-Boson Representation

To handle the Coulomb repulsion U in the Keldysh framework, we use the large-N slave-boson (SBMF) representation [29, 46, 47]: $d_\sigma^\dagger = f_\sigma^\dagger b$ with $b^\dagger b + \sum_\sigma f_\sigma^\dagger f_\sigma = Q$. Here f_σ are spinons, b is the slave boson, and $Q = 1$ for spin 1/2; only the gauge-invariant amplitude $r = |b_c|$ enters observables.

The renormalized level is $\tilde{\epsilon}_\xi = \epsilon_\xi + \delta_c + \delta_\Sigma$ and the gauge-invariant SBMF scale is $\tilde{\beta} = |\beta_0|r$. For the fixed-$W$ small-width path a seed-independent finite-r solution exists at $T_{\rm b} = 10^{-6}$ ($r \simeq 0.0082$ at the core point; at $T_{\rm b} = 0$ the saddle closes analytically, and $r^2\Gamma_0$ equals the core-free scale $T_K^{(0)}$ at the core EP in the stated wide-band asymptotic, Supplemental Material), whereas at $T_{\rm b} = 0.1$ the closure has no finite-r branch, so the displayed spectra use frozen values (Supplemental Material).

B. Saddle-Point Equations and Gauge-Invariant Amplitude

Varying the complete contour action with respect to δ_c gives the constraint $r^2 + n_f = Q$. In the integrated-bath representation, variation with respect to r gives the coherent-channel and complete influence-kernel forces. An explicit-bath hybridization force is an alternative description, not an additional term for that same bath (Supplemental Material).

The resulting resonance reference is $\omega_{\rm res} \simeq \mathrm{Re}\,\tilde{\epsilon}_\xi$ (Supplemental Material).

C. Near-EP screening-scale estimate

The conventional Schrieffer–Wolff estimate in Eq. (28) uses positive real charge gaps and the bath density. No biorthogonal-projector residue enters its exponent; the supporting scale and flow checks are given in the Supplemental Material.

VI. SPECTRAL PROPERTIES AND PHYSICAL REGIMES

Supplemental Fig. S1 displays the dynamical fingerprints of the exact compensated core [panels (a)–(c)] and of the relative gain–loss-only diagnostic limit [panels (d)–(f)]. The representative coefficients place a symmetric local EP pair at $|\beta_0| = 0.5$, accompanied by a condition-number peak and by a change from bounded oscillation to growth of the dominant mode in the compensated representation. Removing the coherent splitting channel [$\Delta_{\rm eff} = 0$, panels

(d)–(f)] removes the coherent local impurity EP mechanism. The remaining relative gain–loss dynamics is retained only as a diagnostic contrast: it should not be interpreted as a separate positive-screening Kondo scale enhancement.

Condition numbers use the right-eigenvector matrix of the full 4×4 kernel with one normalization throughout and serve only to locate eigenvector coalescence.

A. Causal Spectral Concentration and Weak-Flow Control

The complete causal channel density and the channel-resolved weak-coupling flow provide supporting tests of the analytic construction; the full panels and adverse controls are shown in Supplemental Figs. S2 and S3. For $\beta_0 = 0.50$, the active/control ratios are 1.0567 at $g_* = 0.30$, 1.0179 at $g_* = 0.40$, 0.9893 at $g_* = 0.50$, and 0.9166 at $g_* = 1$. In this flow the channel density is the impurity-resolved causal density $(2\pi)^{-1} G^R \Gamma G^A$, not the bath spin density at the impurity site; because the bare exchange is proportional to the identity, all EP dependence of these ratios enters through that density. They are therefore a bounded diagnostic of the impurity resolvent in the stated window and do not support an enhanced asymptotic Kondo temperature.

VII. FLUCTUATION-DISSIPATION RELATION AND SPECTRAL FUNCTIONS

For the supporting Keldysh calculation we assume a common thermal reservoir distribution, $\Sigma^< = if\Gamma$, with consistent preparation of the quadratic effective problem. Stationarity of the rotating-frame Hamiltonian alone does not establish this state assumption. Under this closure, the physical lesser Green's function is fixed by the bath self-energy. The Supplemental Material verifies that the impurity and bath fluctuation–dissipation relation (FDR) ratios follow the same thermal Fermi function on spectral support, with maximum residual 5.61×10^{-14}. This is a consistency check of the specified closure, not a proof of driven thermalization. No mode-by-mode occupation of the non-Hermitian eigenvectors is used in the physical FDR, in the occupations, or in the saddle-point equations.

The physical lesser function is the bath sandwich $G^<_{\rm phys} = G^R \Sigma^<_{\rm bath} G^A = f(G^A - G^R)$, which also satisfies number continuity, and the physical impurity DOS is $\rho_{\rm imp} = -\pi^{-1} {\rm Im}\, {\rm Tr}[P_{\rm imp} G^R] \geq 0$ (Supplemental Material). A signed response of the relative kernel is not a DOS.

VIII. REAL-TIME DYNAMICS

The real-time diagnostic is the complex survival amplitude $O(t) = \langle \tilde{d}_+ | e^{-i\tilde{H}t} | \tilde{d}_+ \rangle$, evaluated for the frozen kernel of Supplemental Fig. S1(c),(f). For the isolated two-component core, the eigenvalue separation is $2s$; the coupled four-component benchmark also has bath-hybridization beating. A defective block produces a $te^{-iE_{\rm EP}t}$ Jordan transient, while a mode with positive imaginary eigenvalue grows in the compensated representation. With $\Delta_{\rm eff} = 0$ [Supplemental Fig. S1(d)–(f)] the isolated block is broken for every $\Gamma_{\rm PT} \neq 0$, so the coherent and relative-decay channels are both needed for the core EP.

IX. CAUSAL SCATTERING AND LINEARIZED CONTACT ALGEBRA

A. Complete scattering and Jordan response

We use the standard Fisher–Lee and Wigner–Smith constructions only as causal scattering diagnostics of the projected kernel [48–50]; they are not introduced here as new formalisms. For the compensated Markov kernel with positive rate matrix $\Gamma = 2(\bar{\Gamma} I - \Gamma_{\rm PT} \sigma_z)$, the Fisher–Lee matrix $S = I - i\Gamma^{1/2} G^R \Gamma^{1/2}$ is unitary in the passive window and the Wigner–Smith matrix $Q = -iS^\dagger \partial_\omega S$ is Hermitian; only basis-invariant quantities (eigenphases, ${\rm Tr}\, Q$) are used (Supplemental Material).

For the compensated unbroken Markov core, $s = \sqrt{\Delta_{\rm eff}^2 - \Gamma_{\rm PT}^2} \in \mathbb{R}$ and the two causal scattering phases give

$$
\begin{aligned}
{\rm Tr}\, Q(\tilde{\epsilon}_\xi) &= \frac{4\bar{\Gamma}}{s^2 + \bar{\Gamma}^2}, \\
\frac{{\rm Tr}\, Q_{\Gamma_{\rm PT}}(\tilde{\epsilon}_\xi)}{{\rm Tr}\, Q_0(\tilde{\epsilon}_\xi)} &= \frac{\Delta_{\rm eff}^2 + \bar{\Gamma}^2}{\Delta_{\rm eff}^2 - \Gamma_{\rm PT}^2 + \bar{\Gamma}^2} > 1.
\end{aligned}
\tag{24}
$$

Thus, at fixed $\Delta_{\rm eff}$ and $\bar{\Gamma}$, the invariant scattering phase density is analytically concentrated at the resonance center as the EP is approached. Its integrated weight remains fixed, and linewidth- or rate-matched adverse controls need not share the peak ordering. We call Eq. (24) path-specific spectral concentration, not an enhanced thermodynamic Kondo temperature or a basis-independent amplification factor.

The cancellation responsible for its regularity is an exact matrix identity. For the traceless core $h = \Delta_{\rm eff}\sigma_x + i\Gamma_{\rm PT}\sigma_z$, $h^2 = s^2 I$, write $a = z - \epsilon_0 - \Sigma_0^R(z)$. Although the individual biorthogonal projectors $P_\pm = (I \pm h/s)/2$ grow as $1/s$, their complete resolvent is

$$G^R(z) = \frac{P_+}{a-s} + \frac{P_-}{a+s} = \frac{aI+h}{a^2-s^2}. \tag{25}$$

At the EP, $h_{\rm EP}^2 = 0$, and $(aI - h_{\rm EP})^{-1} = \frac{I}{a} + \frac{h_{\rm EP}}{a^2}$, so the coalescence produces a finite-coefficient simple-plus-double-pole Jordan response, with no additional $1/s$ normalization multiplier. This cancellation is proved by Eq. (25); it is not inferred from NRG. The separate NRG audit tests the nilpotent residue direction in a spin-invariant, scalar-hybridization control. An exact interacting dimer calculation establishes a nonzero double pole for that restricted control (Supplemental Material); it does not establish protection in the driven SOC bath. The full-bath interaction results are in Secs. III E and III F. At the EP $\mathrm{Tr}\, Q$ is a Lorentzian with $\int d\omega\, \mathrm{Tr}\, Q/(2\pi) = 2$ whose center value $4/\bar{\Gamma}$ equals that of the Hermitian degeneracy. An off-EP control with the same total rate has a larger raw proper-delay peak because one of its poles is slower; normalized by the slowest pole half-width, the largest proper delay at the EP is 1.83 and 1.72 times that of the Hermitian and off-EP controls (Supplemental Material). The proper delay is therefore a coalescence fingerprint, not a Kondo-scale enhancement.

The same cancellation occurs in the finite-U virtual-charge sum. At particle–hole symmetry, with $A = U/2$ and a channel-independent bare vertex W,

$$\begin{aligned} J_{\rm SW}(s) &= 2W^2\left[(AI-h)^{-1} + (AI+h)^{-1}\right] \\ &= \frac{4AW^2}{A^2-s^2} I, \qquad J_{\rm SW}(0) = \frac{8W^2}{U}. \end{aligned} \tag{26}$$

Thus these complete core and virtual-charge expressions contain no additional Petermann or condition-number multiplier.

B. Low-energy contact-algebra and screening scope

The eigenvalues $\mathsf{s}_a = e^{2i\delta_a}$ of the complete unitary scattering matrix give the exact one-particle finite-size condition and phase density

$$\det\left[I - e^{ik\mathcal{L}} S(k)\right] = 0, \qquad \frac{1}{2\pi i}\partial_\omega \ln\det S = \frac{1}{2\pi}\,\mathrm{Tr}\, Q. \tag{27}$$

After equal-velocity branchwise linearization with a constant component matrix and scalar hybridization, the finite-U two-electron contact matrix is the rational Anderson R matrix in a dressed rapidity; it obeys the Yang–Baxter equation, is $GL(2,\mathbb{C})$ covariant, and the corresponding finite-ring boundary problem admits a unipotent twist, with transformed contact terms and operator domain retained (Supplemental Material and Ref. [36]). No full-momentum integrability is claimed.

For comparison with the causal weak-flow calculation, the only screening estimate retained is the conventional frozen Schrieffer–Wolff form

$$\begin{aligned} J_a^{\rm SW} &= 2|W_a|^2\left(\Delta_{0,a}^{-1} + \Delta_{2,a}^{-1}\right), \\ T_{K,a}^{\rm SW(est)} &\simeq D_{{\rm uv},a}\exp\left[-1/(\chi_K \rho_{b,a} J_a^{\rm SW})\right]. \end{aligned} \tag{28}$$

Here $\Delta_{0,a}, \Delta_{2,a} > 0$ are real empty- and double-charge excitation costs in a conventional local-moment reference and $\rho_{b,a}$ is its bath density. The coefficient χ_K is defined by the one-loop convention $dj/d\ell = \chi_K j^2$, with $j = \rho_b J$ and $\ell = \ln(D_{\rm uv}/\Lambda_{\rm run})$ for running cutoff $\Lambda_{\rm run}$. This estimate is not evaluated at the nearly empty EP witness and does not replace a complex charge denominator by its modulus.

X. CONCLUSIONS

The stationary rotating drive controls a family of resonance EPs in the defined finite-band radial SOC impurity model. In the constant-vertex, two-open-channel window it supplies coherent spin-odd coupling and modifies the SOC-induced relative decay. The full-frequency double zero, rank-one kernel and pole exchange establish the coalescence

numerically while the physical decay rates remain positive. This is the principal result: the microscopic controls reach an EP of the causal resonance problem beyond its normalized constant-core approximation.

The interaction calculations probe continuation of that result. The finite-chain self-energy embedding retains the tested topology at $N = 10$ and 12, without establishing convergence of the occupied bath. The continuum KR approximation extends the low-filling EP to $U = 80\bar{\Gamma}_0$ and admits charge-fluctuating EPs at and near mean filling one. Nearby EPs persist beyond folds of the exactly filling-one branches. A local-moment EP and exact thermodynamic interaction survival remain unresolved.

The supporting analyses delimit the interpretation. Complete propagators and the restricted virtual-charge sum cancel divergent individual projectors; the normalized proper delay does not establish enhanced Kondo screening. The scalar-bath Jordan control and equal-velocity contact algebra concern separately specified models. Their symmetry restrictions cannot be imposed on the full SOC self-energy to infer an interacting EP. The next interaction test is a continuum-converged self-energy at an occupied, charge-quenched point, followed by retuning the resonance pair. Pumped screening additionally requires specified reservoirs and consistent retarded and lesser self-energies.

CODE AND DATA AVAILABILITY

The original certificates and finite-chain outputs are archived in Release v2.1.0, doi:10.5281/zenodo.21888925, and the public repository `VMKPHYSMATH/emergent-pt-dirac-impurity` [51]. The revision of 28 October 2026 provides the corrected manuscript, finite-U saddles, filling and bath-convergence checks, supporting symmetry control, validation records and figure sources in the repository's dated revision directory. This revision is distinct from v2.1.0. A matching fixed-version GitHub release and Zenodo deposit will be completed before journal submission; their identifiers will be added after deposition. The all-versions record is doi:10.5281/zenodo.21434682.

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